Sheet 54 Tue 23 Mar. 2027 *Real data: processing a file and filtering (AND, OR, NOT)*

1. Warm-up · Quick In a file of 1 000 rows, 400 satisfy A, 300 satisfy B and 100 satisfy both A and B. How many satisfy "A AND B"? "A OR B"?
2. Core · Method With the same data: how many satisfy "NOT A"? "A AND NOT B"?
3. Core + · Reading code What is the final value of `compte`?

```
compte = 0
for age in [15, 16, 17, 16, 18]:
    if age >= 16 and age < 18:
        compte = compte + 1
```

4. Review · Quick Give the decimal form of $\frac{7}{8}$, then the fraction in lowest terms equal to 0.12.
5. Thread · Formula → table $f(x) = x^2 - 2x$. Complete a table of values for $x \in \{0\,;\,1\,;\,2\,;\,3\}$.

Sheet 55 Thu 25 Mar. 2027 *Critical reading of statements from the media*

1. Warm-up · Quick A 10 % increase on 40: what is the final value?
2. Core · Critical thinking An article claims: "60 % of accidents involve sober drivers". Can we conclude that driving sober is dangerous?
3. Core + · True/false True or false? Justify. "Two smokers out of three are men" means that two men out of three smoke.
4. Review · True/false True or false? Justify. The median of a data set is always equal to its mean.
5. Thread · Reading code What is the value of `s` at the end?

```
s = 0
for i in range(1, 5):
    s = s + i
```

Coordinate geometry: vectors and lines, a synthesis

Sheet 56 Tue 30 Mar. 2027 *Vectors and collinearity: consolidation*

1. Warm-up · Quick $\vec{u}\begin{pmatrix}3\\-2\end{pmatrix}$ and $\vec{v}\begin{pmatrix}-1\\5\end{pmatrix}$. Calculate $\vec{u}+\vec{v}$ and $2\vec{u}$.
2. Core · Method Are the points $A(-1;2)$, $B(1;3)$ and $C(5;5)$ collinear?
3. Core + · Method $A(-1;2)$, $B(3;1)$, $C(4;4)$. Find D such that $ABCD$ is a parallelogram.
4. Review · Quick In a class of 24 pupils, 18 have school lunch. What proportion is this, as a percentage?
5. Thread · Prove Show that $(n+1)^2-(n-1)^2 = 4n$ for every integer n. What can be concluded about the difference of the squares of two consecutive odd numbers (n even)?

Sheet 57 Thu 1st Apr. 2027 *Determinant and equations of lines: consolidation*

1. Warm-up · Quick What is the gradient of the line $3x + 2y - 6 = 0$?
2. Core · Method Equation of the line (AB) with $A(1;2)$ and $B(4;8)$.
3. Core + · Method Are the lines $2x - y + 1 = 0$ and $4x - 2y + 5 = 0$ parallel, coincident or intersecting?
4. Review · Quick Calculate $10^{-2} \times 5 \times 10^{4}$.
5. Thread · Method Solve in $\mathbb{R}$: $7 - 2(x-1) = 3x - 6$.